

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	1	of	17

NPS/003/017 – Technical Specification for a Ground Mounted Distribution Switchgear Remote Control System.

1. Purpose

The purpose of this document is to detail the requirements of NEDL and YEDL to remotely control and monitor 11kV, or 20kV, switchgear in ground mounted distribution substations.

This document supersedes the following document, all copies of which shall be destroyed.

Ref	Version	Title
NONE		

2. Scope

This specification details the requirements for a ground mounted plant Remote Control System in YEDL & NEDL distribution substations. The specification covers the functionality of the units and operational safety requirements, but does not include requirements for the site to base communications.

3. Technical Requirements

3.1 Summary

NEDL & YEDL require the ability to remotely control and monitor between one and three switchgear devices in ground mounted distribution substations.

The communication and logic systems shall all be capable of controlling and monitoring up to three switchgear devices at any site.

The vast majority of substations to be remotely controlled are in an indoor environment, but some are in an outdoor environment.

3.2 Operating Characteristics

The system and all software shall comply fully with IEC 61508 – Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems.

The system shall comply fully with IEC60439 Part 1 - Low Voltage Switchgear and Controlgear Assemblies

The system shall be designed to be failsafe and shall not present risk to any persons, or to any plant and equipment.

The system must incorporate a minimum of 3 seconds delay between subsequent switching operations on the same item of switchgear.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	2	of	17

The time taken from the actuator receiving an operational signal to the point that the switchgear operates and the RTU detects the switchgear change of state, shall not exceed 5 seconds.

Following a successful command/action/report sequence, the system shall be ready and able to begin another command/action/report sequence within 3 seconds of completing the first sequence.

The system shall pick up signals from the volt free contacts on the local FPIs and shall relay these to the Control Centre via the SCADA radio link.

Where latching relays are to be used to control plant and/or equipment, provision shall be made to ensure that no relay shall remain latched for more than 7½ seconds.

The design and installation of the remote control scheme shall ensure that any differences in touch or step potential, particularly due to the passage of fault current, will not affect the switchgear or actuator, or endanger any persons.

3.3 Operating Environment

All of the equipment that forms part of an on-site remote control system shall be designed and proven to work under the following conditions:

Operating ambient temperature range: -25°C to +40°C (+ solar gain of 1,000W/m2)

Relative humidity: Up to 95 %, at 40°C non-condensing (tropicalised).

All of the equipment that forms part of an on-site remote control system shall be designed and proven to comply with the following:

European Community (EC) Electro Magnetic Compatibility directive

(EMC) directive No. 89/336/EEC - Remote control equipment and systems specification BS EN 60870-2-1, Level 4.

European Community (EC) Low Voltage (LVD) directive No. 73/23/EEC - Safety of information technology equipment, including electrical business equipment specification BS EN 60950.

3.4 Remote Control Cubicle

3.4.1 The Cubicle

The remote control cubicle shall be designed to accommodate, as a minimum, the following equipment: Communications Radio, RTU, Local Control Panel and Battery & Charger.

The cubicle and its contents shall comply with IEC60439 Part 1 – Low Voltage Switchgear and Controlgear Assemblies.

IEC60439 (Annex E) - items subject to agreement between manufacturer and user and which are applicable for this application are:

IEC60439 - 6.2.3 – Heavy pollution of the air by salt is likely to occur in coastal regions

IEC60439 - 6.2.4 – Exposure to strong electric and magnetic field is likely

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	3	of	17

IEC60439 - 6.2.5 – Exposure to solar radiation is likely, and this shall be assumed to be 1,000W/m². The temperature rise inside the battery chamber shall be taken account of when calculating battery life.

IEC60439 - 6.2.6 – Attack by fungus and by small creatures is likely

For outdoor use it shall have a minimum IP rating of IP44D to IEC 60529. It shall also meet the requirements of IEC60439 part 5.

For indoor use it shall have a minimum IP rating of IP32 to IEC 60529.

The cubicle shall be proven and certified intruder resistant to a minimum of the UK Building Research Establishment Loss Prevention Standard BRE-LPS1175 level 1.

The cubicle shall be equipped with a hinged, front opening door with padlocking facilities that will accept YEDL or NEDL padlocks. These padlocks have a body up to 63mm square with up to 10mm diameter shackle having clear inside width of 35 mm and a shackle inside length of between 25mm and 45mm. The hole provided for the shackle shall be not less than 11mm diameter.

The cubicle shall have maximum dimensions of 550(h) 500(w) 330(d) mm.

The cubicle shall not require any maintenance for a minimum of 30 years.

The cubicle will be fitted with ventilation compatible with the environment in which it is to be used and the equipment it is to contain, including rechargeable battery use.

To allow for flooding of the substation; the remote control cubicle arrangement shall be designed so that the height to the bottom of the cabinet is 1.2m above floor level.

The remote control cubicle shall be designed so that it can be fitted in at least one of the following three positions:

1. On the switchgear and arranged such that access to the switchgear for normal operation, maintenance and testing of the switchgear is maintained.
2. Mounted on a freestanding support.
3. Wall mounted.

Provision should be made for multicore cable termination for plant cables and the LV auxiliary supply via industry standard plugs and sockets.

A double-ended bulkhead "TNC" type socket shall be provided for radio aerial feeder connection.

A suitably rated earthing point shall be provided on the outside of the cabinet.

The cabinet shall have durable labels, each with the substation name and number permanently fixed to the outside and to the inside, of its door.

Encapsulated schematic and physical layout diagrams shall be provided in every cabinet. These diagrams shall detail the location, type and rating of all fuses, links, terminal blocks, LEDs and external connections.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	4	of	17

3.4.2 Local control operation panel

Facilities for local control of the switchgear shall be provided within the cubicle.

Separate controls shall be provided for 'OPEN' and for 'CLOSE' operations with separate sets of controls shall be provided for each switch. These controls shall all be individually, and independently, lockable, with either YEDL/NEDL safety padlocks, or split rings.

Each pair of controls shall have a label attached with the feeder name of its associated switchgear.

All controls that do not have any associated switchgear, or function, shall have blank feeder name labels. These unassociated controls shall also be obscured from the operator's normal operating position.

Normally, control of the switchgear will be through the RTU remote control interface. To ensure that control is only possible remotely or locally at any one time a single padlockable Control Isolation rotary switch, shall be fitted to the local control panel. This switch shall have only two positions; and these shall be labelled 'Remote' and 'Local'.

The Control Isolation switch shall operate as follows:

The "Local" position shall disable and prevent any control being actioned remotely via the RTU, but shall not disable any remote indication or alarm facilities.

The "Local" position shall enable local control facilities, which shall be independent of the RTU. A 'local selected' status signal shall be presented to the RTU for inclusion into the SCADA host reporting database.

The "Remote" position shall enable controls being actioned remotely via the RTU.

The "Remote" position shall disable and prevent all local control facilities, but shall not disable any local indication or alarm facilities.

A 'remote selected' status signal shall be presented to the RTU for inclusion into the SCADA host reporting database.

An MCB shall be fitted to protect the integrity of the 24V switching supply to the control functions. The MCB shall provide remote indication, raising a remote common alarm when open.

A point of isolation shall be provided to enable isolation of the 24V supply to the actuator(s).

Indication shall be provided in the control cubicle, to replicate the remote common alarm condition. This indication shall be so positioned that it is clearly visible to a local operator, when the cubicle door is opened.

3.4.3 RTU

The RTU, and its software, shall comply fully with IEC 61508 – Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems

The cubicle shall be fitted with a remote control interface. This interface shall be a versatile, high performance RTU device with intelligent programming capability to enable communication with the Control System located at any CE Electric's Control Centre

Loss of battery supply, or low battery volts, shall not cause any loss of stored data within the RTU.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	5	of	17

RTU Programming capability

The RTU shall be equipped for the programming of and/or the implementation of specific, locally derived, control sequences.

The RTU shall be supplied ready to accept downloaded generic configurations that are prepared utilising an external workbench utility.

RTU Configuration and supporting utilities

The configuration utility shall be capable of running on a standard IBM compatible laptop PC.

A Windows based RTU configuration software utility, which allows modifications to the RTU database configuration shall be provided. This software utility shall be compatible with MS Windows 95, 2000, NT and XP.

The configuration utility or a separate monitoring utility shall enable connection to the RTU to provide monitoring of the status of the plant I/O.

The software shall provide direct access to the RTU database and allow confirmation of the current status of all I/O points within the RTU.

These utilities shall be capable of saving and/or loading the configuration and/or data files to disc.

These utilities shall be capable of saving a plain text file formatted version of the RTU settings.

RTU Status inputs

The RTU shall include the following I/O:

16 local inputs to include support of the LOCAL/REMOTE switch, and battery monitoring and actuator supply off.

All inputs shall be electrically isolated from the processor electronics

Secure Control Outputs. Each channel will employ 2-bit control (Open and Close per channel) isolated from all other Plant I/O. All outputs shall be capable of High security Select-Checkback-Execute sequence operation.

3 Control Outputs for switchgear control

- 1 Control output for 12V DC radio power management

- 1 Control per Fault Passage Indicator (FPI) reset

- 1 dummy control internal to the RTU.

RTU Serial Interfaces

Configurable serial interfaces shall be provided and available as follows:

- Two RS232 or RS485 serial interfaces for host communications or master port interface to slave RTU or IEDs to be equipped with Modem control signals

- Three RS232 or RS485 serial interfaces for slave port interface to RTUs or IEDs

- An RS232 for configuration/diagnostic tool

Document reference		NPS/003/017				
Version:-	1.0	Date of Issue:-	March 06	Page	6	of 17

RS485 serial ports shall provide Input/output isolation of 1.5 kV rms / 2.1 kV DC between the input supply and all other card circuits.

RTU database

Plant Functionality

Function	Control	Indication	Alarm	Analogue	Comment
Bay 1 Open/ Close CB/Switch	Yes	Yes			
Bay 1 Protection/FFI Operated			Yes		Where available
Bay 1 Operations counter				Yes	
Bay 2 Open/ Close CB/Switch	Yes	Yes			
Bay 2 Protection/FFI Operated			Yes		Where available
Bay 2 Operations counter				Yes	
Bay 3 Open/ Close CB/Switch	Yes	Yes			
Bay 3 Protection/FFI Operated			Yes		Where available
Bay 3 Operations counter				Yes	
DCB Open/Close	Yes	Yes			
Local/Remote operation		Yes			Common for all three bays
230V Supply Fail (delayed by 120 seconds)			Yes		
Common Alarm			Yes		
Total	4	5	5	3	Total: 17

Digital Inputs

Index	Name	Classes	Comment
0	Bay 1 Switchgear Tripped	1	
1	Bay 1 Switchgear Closed	1	
2	Bay 2 Switchgear Tripped	1	
3	Bay 2 Switchgear Closed	1	
4	Bay 3 Switchgear Tripped	1	
5	Bay 3 Switchgear Closed	1	
9	Mode (Local/Remote)	1	
10	Bay 1 Protection/FFI operated	1	
11	Bay 2 Protection/FFI operated	1	
12	Bay 3 Protection/FFI operated	1	
13	Common Alarm	1	To indicative the failure of any sub system such as: Battery fail Local communications fail RTU problem
16	Dummy Circuit Breaker Closed	1	
17	Auxiliary Supply Fail (delayed)	0	The class should be configurable in the RTU configuration tool

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	7	of	17

Analogue Inputs

Index	Name	Class	Comment
0	Bay 1 Operations counter	0	
1	Bay 2 Operations counter	0	
2	Bay 3 Operations counter	0	
3	Voltage (i)	0	

(i) Auxiliary supply referred to 11 kV. Scaling factor 1:1 at 11,000V

Binary Outputs

Index	Name	Comment
0	Bay 1 Open/Close CB/Switch	
1	Bay 2 Open/Close CB/Switch	
2	Bay 3 Open/Close CB/Switch	
3	Reset fault flags/FFI	Where available, dependent on relays/FFI.
6	DCB Open/Close	

DNP3

All RTUs shall have, at least, DNP3 2002 L2 approval.

All binary outputs shall be Direct Operate.

All DNP3 internal indication bits shall be supported.

All controls shall use the Control Relay Output Block (object 12) and be confirmed by a Binary Input Change object.

Control Operations shall be selectable as:

Select Before Operate

Direct Operate

All points shall be capable of supporting the following controls:

Pulse On

Pulse Off

Latch On

Latch Off

In addition all plant controls shall support Trip/Close

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	8	of	17

Minimum message set

Communications to remote distribution equipment shall show the following identified message types:

Datatype	Object	Variation	Unsolicited	Function Codes		Plant Function	Comment
				To RTU	From RTU		
Binary IP	1	1 2		1, 22	129	All points	Without Status Flags Include status flags
Binary IP Change with Time	2	1	Yes	1	129, 130	Plant operations, protection operations, alarms	All events to need to include time stamps. See comment below. Without status flags.
		2					All events to need to include time stamps. See comment below. Include status flags.
Binary OP Status	1 0	2		1	N/A	All controls	Required for Class 0 polling to indicate control status
Control Relay OP block	1 2	1		3,4,5	129	All controls	
16 bit Analogue IP	3 0	2 4		1, 22	129	Phase voltage, line current	Include status flags Without status flags
16 bit Analogue IP Change with Time	3 2	2 4	Yes	1	129, 130	Phase voltage, line current	Without status flags Include status flags
Time and date	5 0	1		2	129	Time synchronisation	Comms check if application confirm can be generated
Time delay fine	5 2	2		N/A	N/A	Used to measure Comms latency	
Class data 0	6 0	1		1, 20, 21	N/A	Class requests	
Class data 1	6 0	2		1, 20, 21	N/A	Class requests	
Class data 2	6 0	3		1, 20, 21	N/A	Class requests	
Class data 3	6 0	4		1, 20, 21	N/A	Class requests	
Internal indications	8 0	1		2	N/A	RTU and event buffer status	

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	9	of	17

The following function codes shall be supported:

From the Control System:

Function Code	Description	Comment
1	Read	
2	Write	
3	Select	
4	Operate	
5	Direct Operate	With Acknowledgement
20	Enable Unsolicited Messages	
21	Disable Unsolicited Messages	
22	Assign Class	
23	Delay Measurement	

From the RTU:

Function Code	Description	Comment
129	Response	
130	Unsolicited Message	

All messages require an application confirm from the control system. Data Links are disabled.

All controls shall be performed with the Control Relay Output Block, object 12 variation 1.

All controls shall be Direct Operate.

Confirmation of controls shall be via an unsolicited change in the relevant binary IP.

All analogues shall be 16 bit.

Time and date synchronisation shall use delay measurement.

The reporting of all plant statuses shall be single bit.

Binary and Analogue objects with and without status flags.

All status changes assigned to class 1 shall be reported immediately.

All unsolicited events shall be date and time stamped in the RTU when the event occurs.

Upon start-up and reset; RTUs shall not send unsolicited messages (apart from those used to indicate to the control system the availability of the RTU) until the control system enables them.

3.5 Energy Monitoring and Control

The RTU shall monitor the LV supply to the cubicle and shall initiate an instantaneous suppressed LV Supply Fail indication and a Delayed Supply Fail Indication, should the supply fail persist for longer than a

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	10	of	17

defined time, which shall be configurable within the RTU. This time delay shall be user configurable up to a minimum of 3 hrs, in 1-second steps.

Upon the failure of the auxiliary supply a timer shall be initiated and, after a configurable time (up to 24 hours settable in 1 minute intervals), the power to the radio shall be automatically disconnected.

The Delayed Supply Fail Indication shall be reset after the instantaneous indication has been reset for 20 seconds.

Upon the restoration of the auxiliary supply, the power to the radio shall be immediately, automatically reinstated.

3.6 Connecting Cables

Volt Drop

The power supply connection from the control cabinet to the actuator shall be designed to minimise volt drop and shall, in any case, limit this drop to 1.5V.

Physical Protection

The connections between the RTU and the actuator and/or switchgear shall be provided by suitably armoured cables, or protected in suitable, flexible conduits.

All cabling between the RTU, actuators and switchgear shall be secured and protected. All cables shall be routed so as not to interfere with the normal operational and maintenance requirements of the switchgear.

Terminations

Where preformed cables are supplied they shall have a plug/socket arrangement at each end. The plug and socket shall be of standard design such that an RTU can be connected to any actuator with the same preformed cable. I.e. if core 1 and 2 is assigned for the actuator close command for one design of actuator they shall have the same function for all actuators.

Where more than one cable is required, to connect between RTU and switchgear and/or the actuator, the cables, plugs and sockets shall be clearly labelled to ensure they can be readily identified and installed in the correct orientation.

Signals

The cables shall provide signals for:

- 1 Open and 1 Close control per switch
- 1 Earth Fault Reset control per switch
- 1 Open/Close double bit indication per switch
- 1 single bit earth fault indication per switch
- 24V for switchgear use
- 4 spare connections for serial connections to IEDs
- Common

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	11	of	17

3.7 Communications

The RTU shall include a minimum of five serial ports: at least two shall be suitable for host communication and at least 3 shall be suitable for slave communications.

The communication protocols shall be configurable by means of a software configuration program.

The RTU shall support both DNP3 Level 2 and Modbus communications protocol to host or slave. The protocol used to communicate between Control centre and all remote site equipment RTUs shall be a minimum of DNP3 L2. All supplied RTUs shall hold a minimum of a DNP3 L2 compliance certificate.

3.7.1 Communication Media

A wide range of communications media shall be supported. Provision shall also be provided for compatibility with any/all of the following media:

PMR radio (MPT1327)

UHF/VHF low power radio

Paknet, or GSM, cellular networks

PSTN dial up

Pilot wire

De-regulated low power (MPT1329 including repeaters if required)

Regulated radio (MPT1411)

Fibre Optic

Low Earth Orbit (LEO) Satellite

The Control Cubicle shall be able to house suitable radio sets, modems, or equivalent terminal equipment, for use with any one of the above.

3.7.2 Communication Details

The associated communications radio shall be accommodated within the control cubicle.

The communications radio aerial connection shall be mounted, and connected, via a bulkhead fitting, typically either a TNC-TNC on both sides or an N type-TNC, through the Control Cubicle, to the radio (having typically a BNC, or N type, flying lead as an integral part of the radio).

A fuse protected 12 V DC power supply, switched via a secure RTU control signal, shall connect the internal RTU power system to the radio.

A data cable, with suitable terminations, shall connect the radio, typically having a Molex plug, or screw terminals, and the RTU host communications port.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	12	of	17

The radio network may at times be busy performing internal functions and not be ready to accept a DNP transaction for transmission across the network. To cater for these situations a method of communications flow control between the RTU and the radio shall be in place.

The RTU shall hold off communication until the Clear To Send (CTS) signal from the radio is active.

The serial port hardware handshaking shall work using levels rather than edges in the detection of the CTS state.

The RTU shall have a user configurable delay before a message is sent after the RTS is raised and CTS is high. This delay resolution shall not be greater than 10mSec.

3.8 Power Supplies

The incoming power supply unit will be 230V AC +/-10%. The remote control system shall be capable of functioning fully and normally across this input range.

The control cabinet shall be connected to the 13A supply, using suitably rated and mechanically protected cable and/or conduit.

The incoming supply, battery charger and battery shall be rated to provide power to the RTU, communications and the actuator(s).

3.8.1 The Battery

The battery and the overall remote control system shall be designed to be sufficiently robust to prevent spurious low volt alarms resulting from normal operation of the remote control system.

The battery unit shall provide a minimum of 24AH capacity.

A 12V DC supply shall be provided to power the radio. This supply to the radio shall be protected from overloads above 7A and shall be controllable from the RTU.

The batteries shall have an actual 10-year operational life, taking into account fully the predicted site conditions.

Following loss of mains supply for a continuous 18 hour period, during which time ambient temperatures are minus 25°C, the battery shall be capable of powering the communications and RTU.

At the end of this 18 hour period, the battery shall be capable of powering the communications, RTU and actuator to carry out 6, consecutive, Open/Close (3 open + 3 close) operations of the switchgear, also at an ambient temperature of minus 25°C.

3.8.2 Battery Charger

The battery charger shall have a minimum life of 20 years, taking into account fully the predicted site conditions.

The battery charger shall be temperature compensated.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	13	of	17

The battery charger shall recharge the batteries, from discharged to fully charged, in a maximum of 6 hours. This requirement assumes that the RTU, communications and FPI are connected and functioning during this charging time, but that no actuator operation is required.

3.8.3 Battery Protection, Monitoring and Testing

The battery shall be protected against overload and against short circuit.

The system shall load test the battery every 24 hours and shall initiate an alarm if a problem is found.

The voltage at the battery terminals shall be monitored continuously.

A battery volts low alarm shall be initiated, via the RTU, if the battery terminal volts drop below 22V, for more than 10 seconds.

The battery shall be protected against deep discharge. The battery shall be disconnected immediately if its terminal volts drop below 20V, for more than 10 seconds.

3.9 Actuators

Actuators shall operate the switchgear in a safe and effective manner.

There shall be provision for manual operation of the switchgear.

The actuator must not interfere with visual confirmation of the position of the switch that it is controlling, by allowing an unobstructed view of the switchgear's original On/Off indication.

The indication of the position of the switch shall be derived from the switch mechanism and not from the actuator.

The actuator shall be designed so as to allow compliance with CE Electric's current operational policies. There shall be provision to allow a YEDL or NEDL operational, or safety, padlock to be applied to prevent any operation (Remotely, Manually or Locally) of the switchgear.

The actuator shall have been type tested over a minimum 1000 full cycle operations and have a life expectancy of 15 years minimum without maintenance, taking into account fully the predicted site conditions.

The actuator shall complete its operation within 5 seconds of receiving an initiation signal.

If the actuator fails, or loses power, partway through an operation, it shall be capable of dissipating any stored energy, or there shall be a safe, documented procedure in place to allow the manual removal of the actuator.

All actuators shall be permanently marked with a unique manufacturer's serial number.

A full set of instructions detailing the procedure for the fitting and removal of the actuator shall be located on site within the control cubical.

The actuator shall be software and/or logic free.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	14	of	17

All actuator fastenings and connections shall be categorised as either:

- (a) required for use in normal switchgear operation in which case they shall not require the use of any tools.
- (b) not required for use in normal switchgear operation in which case they shall require the use of a tool to unfasten/unfasten or disconnect/reconnect; these fastenings shall not utilise a simple slotted screw head, but shall require the use of an Allen key, splined drive, or similar.

3.9.1 Fitting

The actuator shall be of a non-intrusive design that can be fitted to the switchgear with the equipment live and in service position.

The fitting of the actuator shall not compromise the integrity of the switchgear that it is to operate.

The remote control arrangement shall not impede access to the HV cable boxes, HV test orifices, the switch position indication or any associated Fault Passage Indicator.

The actuator shall be able to be fitted to a switch in the middle, or at either end of a switchboard in a substation. The actuator shall not interfere with the operation of any adjacent switchgear.

3.9.2 Removal

The actuator shall be designed in such a manner that it can be removed easily from the switchgear. The removal shall require no special tools and shall be a single person operation.

3.9.2 Refitting

The system shall be designed in such a manner that, following the temporary disconnection of the actuator as part of the normal switching sequence; the actuator can only be refitted so that the actuator state and the RTU state correspond to the switchgear state.

The refitting shall require no special tools and shall be a single person operation.

To give operator confidence; the RTU shall present indication to confirm the status of the switchgear/actuator, as interpreted by RTU. This indication shall be so positioned that it is clearly visible to a local operator, when the cubicle door is opened.

3.10 Fault Passage Indicator

The outputs from the FPI volt free contacts shall be connected to the remote control system.

Where an FPI has two volt free contacts these outputs shall be paralleled up within the control cubicle, before being connected into the remote control scheme.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	15	of	17

4. References

4.1 External Documentation

Equipment shall comply with the relevant International Standards, British Standard Specifications and all relevant Electricity Association Technical Specifications (EATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

Reference	Title
IEC 61508	Functional Safety of Electrical/Electronic/Programmable Electronic Safety-Related Systems
IEC60439 Part 1	Low Voltage Switchgear and Controlgear Assemblies
IEC 60529	Degrees of Protection Provided by Enclosures
The UK Building Research Establishment Loss Prevention Standard BRE-LPS1175	Requirements and Testing Procedures for the LPCB Approval and Listing of Burglary Resistant Building Components, Strongpoints and Security Enclosures.
European Community (EC) directive No. 89/336/EEC	Electro Magnetic Compatibility (EMC) Directive.
BS EN 60870-2-1	Telecontrol Equipment and Systems Part 2: Operating Conditions Section 1: Power Supply and Electromagnetic Compatibility
European Community (EC) directive No. 73/23/EEC	Low Voltage (LVD) Directive
BS EN 60950	Safety of Information Technology Equipment, Including Electrical Business Equipment
BS EN ISO 9001	Quality Management Systems - Requirements
BS EN ISO 14001	Environmental management systems Requirements with guidance for use

The supplier shall provide with the tender full technical details of the equipment offered and shall indicate any divergence from these standards or specifications.

5. Definitions

Term	Definition
Remote Control Cubicle	The cubicle housing items including: the radio, the RTU, the battery & charger, the Local Control Panel, etc.
RTU	A Remote Terminal Unit. - An item of equipment typically used as an interface between a remote device requiring both analog or digital I/O and a communications device.
DNP3	Distributed Network Protocol - A multi-layered

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	16	of	17

	communications protocol used in SCADA Communications systems.
SCADA	Supervisory, Control and Data Acquisition system. - A SCADA system usually consists of multiple RTUs distributed across a wide area, communicating to a central computer running a network management system.
Manual Operation	Whereby an operator on site manually operates the switchgear
Local Operation	Whereby an operator on site operates the switchgear, via on-site controls that are part of the remote control scheme.
Remote Operation	Whereby a command signal is sent, usually from a Control Centre, to site and the switchgear is operated by the remote control scheme.

Document reference		NPS/003/017					
Version:-	1.0	Date of Issue:-	March 06	Page	17	of	17

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
D. Blackledge	Plant & Switchgear Standards Manager	D. Blackledge	06/03/2006

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G. Bolden	Purchasing Manager	G. Bolden	06/03/2006

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D. Gazda	Standards Co-ordinator	D. Gazda	07/03/2006

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	31/03/2006

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C. Holdsworth	Standards and Assessment Manager	C. Holdsworth	07/03/2006

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C. Morgan	Head of Procurement	C. Morgan	23/03/2006